

```
In [1]: import pandas as pd
from sklearn.metrics import confusion_matrix, classification_report, accuracy_score
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
```

```
In [2]: df = pd.read_csv('Datasets/Iris.csv')
```

```
In [3]: df
```

```
Out[3]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

```
In [4]: df.isnull().sum()
```

```
Out[4]: Id          0
SepalLengthCm      0
SepalWidthCm       0
PetalLengthCm      0
PetalWidthCm       0
Species           0
dtype: int64
```

```
In [5]: label_encoder = LabelEncoder()
```

```
In [6]: df['Species'] = label_encoder.fit_transform(df['Species'])
```

```
In [7]: df
```

Out[7]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	0
1	2	4.9	3.0	1.4	0.2	0
2	3	4.7	3.2	1.3	0.2	0
3	4	4.6	3.1	1.5	0.2	0
4	5	5.0	3.6	1.4	0.2	0
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	2
146	147	6.3	2.5	5.0	1.9	2
147	148	6.5	3.0	5.2	2.0	2
148	149	6.2	3.4	5.4	2.3	2
149	150	5.9	3.0	5.1	1.8	2

150 rows × 6 columns

In [8]:

x = df.drop('Species', axis=1)

In [9]:

x

Out[9]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	1	5.1	3.5	1.4	0.2
1	2	4.9	3.0	1.4	0.2
2	3	4.7	3.2	1.3	0.2
3	4	4.6	3.1	1.5	0.2
4	5	5.0	3.6	1.4	0.2
...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3
146	147	6.3	2.5	5.0	1.9
147	148	6.5	3.0	5.2	2.0
148	149	6.2	3.4	5.4	2.3
149	150	5.9	3.0	5.1	1.8

150 rows × 5 columns

In [10]:

y = df.Species

In [11]:

y

```
Out[11]: 0      0
         1      0
         2      0
         3      0
         4      0
         ..
        145    2
        146    2
        147    2
        148    2
        149    2
Name: Species, Length: 150, dtype: int64
```

```
In [12]: x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=0)
         gaussian = GaussianNB()
         gaussian.fit(x_train, y_train)
```

```
Out[12]: GaussianNB ⓘ ?
         ► Parameters
```

```
In [13]: y_pred = gaussian.predict(x_test)
```

```
In [14]: matrix = confusion_matrix(y_test, y_pred)
```

```
In [15]: matrix
```

```
Out[15]: array([[11,  0,  0],
                [ 0, 13,  0],
                [ 0,  0,  6]])
```

```
In [16]: print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	11
1	1.00	1.00	1.00	13
2	1.00	1.00	1.00	6
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

```
In [17]: accuracy = accuracy_score(y_test, y_pred)
```

```
In [18]: accuracy
```

```
Out[18]: 1.0
```

```
In [20]: print('True Positive:', matrix[0][0])
         print('True Negative:', matrix[1][1])
         print('False Positive:', matrix[0][1])
         print('False Negative:', matrix[1][0])
         precision = matrix[0][0]/(matrix[0][0]+matrix[0][1])
         print('Precision:', precision)
```

True Positive: 11
True Negative: 13
False Positive: 0
False Negative: 0
Precision: 1.0